

The Privacy Subsidy: Kyle’s λ under Noise-Perturbed Order-Flow Observation

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Abstract. Privacy-preserving cryptocurrency exchanges alter what the pricing mechanism observes about order flow. We derive the unique linear Kyle equilibrium when a committed Bayesian market maker observes order flow perturbed by independent Gaussian privacy noise. The price-impact coefficient and informed-trader strategy rescale by reciprocal factors of the privacy parameter (one down, one up), so their product is invariant. A welfare decomposition then identifies a closed-form per-period transfer from the protocol’s LP pool to traders — the *privacy subsidy*, the break-even fee any privacy-aggregated exchange must charge. The result is the single-period closed-form privacy-noise analog of Loss-Versus-Rebalancing [13]. The primary application is shielded AMMs with explicit additive-noise injection (e.g., differential privacy); related designs (batched swaps, sealed-bid auctions, oracle-pegged crossings) require separate frameworks that we leave to future work.

Keywords: Market microstructure · Kyle equilibrium · ZK exchanges
· Adverse selection

1 Introduction

Privacy-preserving exchange designs are an increasingly common architecture in cryptocurrency markets. Renegade matches orders via multi-party computation under zero-knowledge proofs [2]; Penumbra batches swaps and reveals only batch totals on-chain [14]; Suave-style order-flow auctions seal individual bids until builder selection [8]; shielded variants of constant-function automated market makers inject privacy noise into the observable reserve state before on-chain price updates. These designs all alter what the liquidity-providing role — whether an LP pool, an arbitrageur, or a smart-contract pricing rule — observes about order flow.

Classical microstructure theory [11] gives the equilibrium price-impact coefficient λ and informed-trader strategy β in closed form when the market maker observes the full aggregate flow $y = x + u$. These results do not, however, characterize the *welfare* consequences when the market maker prices on a noise-perturbed signal $\tilde{y} = y + \varepsilon$, as arises naturally in privacy-aggregated exchange designs.

Contributions. This paper provides three results, each of which is absent in or distinct from textbook Kyle.

1. A closed-form linear Kyle equilibrium under MM observation $\tilde{y} = y + \varepsilon$ with Gaussian privacy noise $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ independent of (v, u) , under a *committed Bayesian* MM pricing rule: the maker prices at the posterior mean of the coarse signal, which is the competitive price conditional on that signal. The only departure from textbook Kyle is that the maker conditions on $\tilde{y}$ rather than the exact flow y , so it does not break even against the real flow (Remark 2); the equilibrium recovers Kyle exactly in the $\sigma_\varepsilon = 0$ limit. The unique price-impact coefficient is $\lambda = \sigma_v / (2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2})$ with informed-trader linear coefficient $\beta = \sqrt{\sigma_u^2 + \sigma_\varepsilon^2} / \sigma_v$ (Theorem 1).
2. A welfare decomposition under this equilibrium, identifying a per-period transfer $|\pi_M| = \sigma_v \sigma_\varepsilon^2 / (2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2})$ from the protocol/LP side to traders — the *privacy subsidy* (Theorem 2). This quantity is identically zero in textbook Kyle, where the market maker observes order flow exactly; it turns positive once the privacy noise makes the observed signal strictly coarser than the executed flow, so that no price rule can be simultaneously informationally efficient and zero-profit against the real flow (Remark 2). The subsidy is the break-even fee that any privacy-aggregated exchange must charge to compensate its liquidity layer.
3. A primary mapping from σ_ε to shielded AMMs with explicit additive-noise (differential-privacy) injection, yielding a per-design break-even fee. We also delineate the boundary of applicability: Penumbra-style batched swaps reduce to textbook Kyle with rescaled noise-trader variance (and no privacy subsidy under Bayesian MM); Suave-style sealed-bid order-flow auctions sit in LVR’s temporal-asymmetry regime; oracle-pegged crossings such as Renegade fall outside the Kyle framework. These three nearby designs require separate frameworks and are left to future work.

This paper analyzes the equilibrium consequences of privacy in exchange design, treating ZK primitives as a black box producing $\tilde{y}$; protocol-level soundness/completeness analysis is out of scope.

Positioning. On the microstructure side, the result isolates a closed-form welfare quantity (the privacy subsidy) under additive privacy noise on the maker’s flow signal; the equilibrium (λ, β) itself is textbook Kyle under the substitution $\sigma_u^2 \mapsto \sigma_u^2 + \sigma_\varepsilon^2$, so the novelty lies in the welfare object rather than the equilibrium. The closest microstructure precedent on the observability axis is the strand on insider disclosure [10], where the *trader’s* actions become more observable — our case moves the imperfection to the *market maker’s* signal instead. On the decentralized-finance side, the privacy subsidy is the single-period closed-form analog of Loss-Versus-Rebalancing identified by Milionis et al. [13]: LVR isolates the cost to LPs from stale prices being picked off by arbitrageurs with external price information, while our privacy subsidy isolates the cost from privacy noise obscuring the MM’s flow observation. The two are closed-form welfare quantities

of the same family; their combination in a continuous-time AMM with privacy is left to future work (Section 7).

Roadmap. Section 2 surveys related work along three strands. Section 3 sets up the model and equilibrium concept. Section 4 states and proves the main equilibrium theorem. Section 5 derives the welfare decomposition and identifies the privacy subsidy. Section 6 maps the result to canonical zero-knowledge exchange designs. Section 7 discusses the connection to LVR. Section 8 concludes.

2 Related work

2.1 AMM adverse-selection (closest)

Milionis et al. [13] introduce *Loss-Versus-Rebalancing* (LVR): a closed-form continuous-time measure of the cost AMM liquidity providers incur when stale prices are picked off by better-informed arbitrageurs. The two costs differ in source: LVR’s arbitrageurs read an exogenous reference price; our informed trader is the classical Kyle insider, and the cost arises from the AMM’s noisy observation of flow.

Routledge, Shen, and Zetlin-Jones [16] characterize optimal liquidity provision in an AMM and study how price impact depends on trade size and on the dynamics of liquidity provision. Milionis, Moallemi, and Roughgarden [12] cast the profit-maximizing strategy of a monopolist liquidity provider into a Bayesian belief-inference framework with a Myersonian mechanism-design interpretation. Brahma et al. [4] developed a sequential Bayesian-update market maker for binary-outcome prediction markets; that line is structurally related to our committed-Bayesian-AMM rule (Bayesian posterior pricing without a zero-profit constraint), although the model class differs.

2.2 Kyle-style microstructure

Kyle’s [11] single-period model with a risk-neutral monopolist informed trader, an exogenous noise-trader pool, and a competitive risk-neutral market maker is the framework we extend. Foster and Viswanathan [9] characterize the unique linear equilibrium when multiple informed traders receive imperfectly correlated signals. Huddart, Hughes, and Levine [10] modify Kyle by adding mandatory ex-post disclosure of the informed trader’s transactions; this accelerates price discovery and reduces insider profits — essentially the dual of our setup, in which it is the *market maker’s* signal rather than the trader’s identity that becomes opaque. Chhaibi, Ekren, and Noh [6] study a Gaussian Kyle equilibrium with a risk-averse informed trader holding an imperfect signal of the terminal value; the imperfection is on the trader side, not the MM side. Non-fiduciary MMs [1] capture rents through fees; our market maker is instead constrained to a Bayesian rule and *bears* the cost. Viswanathan and Xing [17] analyze informed-trader information acquisition at entropy cost; another direction of imperfection

distinct from ours. A separate strand lets the market maker condition on a noisy observation of the *asset value* alongside the order flow: Qiu and Zhou [15] solve such a continuous-time Kyle model under partial maker observation. Our coarsening is an explicit additive privacy channel on the flow, and our contribution is the closed-form welfare subsidy it induces under committed Bayesian pricing, not the imperfect-observation equilibrium itself.

2.3 Dark-pool and privacy-preserving market structure

Zhu [19] and Buti, Rindi, and Werner [5] analyze dark-pool trading as a routing equilibrium, in which informed and uninformed orders self-select between a lit market and a dark pool based on execution risk. The mechanism is *routing-based segmentation*, which is mathematically distinct from σ -algebra coarsening of a single market’s signal. Bergemann and Morris [3] give the general information-design framework that subsumes σ -algebra coarsening as a special case; we apply it implicitly. Zhang et al. [18] analyze maximal extractable value in batch-auction designs (including a discussion of Penumbra), showing that block-builder reordering of batch contents can still extract MEV; their framework is combinatorial-market (Fisher and Arrow–Debreu) and complementary to ours.

For application targets, we read the Renegade [2] and Penumbra [14] protocol specifications directly; we discuss applicability in Section 6.

3 Model setup

3.1 Primitives

Let $(\Omega, \mathcal{F}, P)$ be a probability space supporting the following independent Gaussian random variables:

$$\begin{aligned} v &\sim N(p_0, \sigma_v^2), \quad \sigma_v > 0, \\ u &\sim N(0, \sigma_u^2), \quad \sigma_u > 0, \quad u \perp\!\!\!\perp v, \\ \varepsilon &\sim N(0, \sigma_\varepsilon^2), \quad \sigma_\varepsilon \geq 0, \quad \varepsilon \perp\!\!\!\perp (v, u). \end{aligned}$$

The random variable v is the asset’s terminal value and p_0 the common prior mean. u is the aggregate flow of uninformed (noise) traders, independent of v . ε is privacy noise introduced by the exchange mechanism (see Section 6 for the realization in specific zero-knowledge designs).

3.2 Players and strategies

There is one *informed trader* (the Kyle insider) who observes v at time 0 and submits an order of size x . We restrict attention to linear strategies $x = \beta(v - p_0)$ for $\beta \in \mathbb{R}_{>0}$, to be determined in equilibrium.

The *noise-trader aggregate* contributes order size u as above. Total order flow is

$$y = x + u.$$

The *market maker* (MM) does not observe y directly. Instead, the MM observes the privacy-noisy signal

$$\tilde{y} = y + \varepsilon.$$

The privacy noise ε perturbs only the signal on which the MM prices (for instance, the observed reserve state); the protocol still settles the true net order $y = x + u$ at the resulting price. This signal-versus-settlement gap is what makes the MM's expected P&L against the real flow nonzero (Section 5). The MM is a *committed Bayesian AMM*: a smart-contract pricing rule that mechanically computes

$$p(\tilde{y}) = \mathbb{E}[v \mid \tilde{y}]$$

which is the competitive price conditional on $\tilde{y}$; the maker breaks even on its own signal but not against the real flow y , which it cannot observe (Remark 2). In a committed smart-contract setting the rule is fixed by the mechanism, so the maker cannot re-optimize away this loss, and the expected loss against the real flow is absorbed by the protocol's LP pool. We motivate the committed-rule setting in Section 3.4.

3.3 Equilibrium concept

We seek a linear equilibrium of the form

$$x = \beta(v - p_0), \quad p(\tilde{y}) = p_0 + \lambda\tilde{y},$$

where $(\beta, \lambda) \in \mathbb{R}_{\geq 0}^2$ satisfy:

1. *Bayesian rationality of pricing*: $p(\tilde{y}) = \mathbb{E}[v \mid \tilde{y}]$ under the conjectured strategy β .
2. *Best response of informed trader*: $\beta(v - p_0)$ maximizes the informed trader's expected profit $\mathbb{E}[(v - p(\tilde{y})) \cdot x \mid v]$, given λ .

3.4 The committed-Bayesian-AMM framing

The maker's restriction to the coarse signal $\tilde{y}$, not any departure from competitive pricing, is the modeling feature distinguishing our setup from textbook Kyle (Remark 2). We adopt this in order to match the design of contemporary on-chain pricing mechanisms, in which:

- the pricing rule is encoded in a smart contract and is mechanically applied to the observable input;
- the input is privacy-noised by construction (e.g., via differential privacy);
- the resulting expected loss is borne by liquidity providers or the protocol treasury, and is recouped through trading fees.

This structure parallels the LVR analysis of automated market makers in Milionis et al. [13], where the MM's pricing rule is the constant-function curve and the cost is borne by LPs. We return to this connection in Section 7.

Idealized benchmark, not literal model. The committed-Bayesian-AMM is an idealized *normative* benchmark: what an informationally optimal smart-contract pricing rule would look like under additive privacy noise. The closest real-world instantiation is a constant-function AMM augmented with differential-privacy noise injection, treated in detail in Section 6. The benchmark’s value is to isolate the welfare cost of additive privacy noise, yielding a closed-form quantity that practical mechanisms in this class must internalize at the fee level.

3.5 Other modeling choices

Four further assumptions warrant brief comment. *Linearity.* We restrict to $x = \beta(v - p_0)$, Kyle’s standard restriction (see Kyle [11], §3); a formal treatment of nonlinear equilibria under privacy noise is outside our scope. *Independence* $\varepsilon \perp\!\!\!\perp (v, u)$. Adaptive privacy mechanisms whose noise correlates with the flow it perturbs are an interesting separate problem; our result establishes the unconditional baseline. *Gaussianity.* The closed-form subsidy is Gaussian-specific; the qualitative existence of a positive privacy-induced LP loss generalizes to other tractable distributions. *Endogenous* σ_ε . A privacy-utility tradeoff that optimizes σ_ε under additional constraints is a natural extension we do not pursue here.

4 Equilibrium under privacy-noisy observation

This section contains the paper’s main equilibrium result.

Theorem 1 (Linear equilibrium under privacy-noisy observation). *Fix $\sigma_v, \sigma_u > 0$ and $\sigma_\varepsilon \geq 0$. The unique linear equilibrium of the model of Section 3 has*

$$\lambda = \frac{\sigma_v}{2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2}}, \quad \beta = \frac{\sqrt{\sigma_u^2 + \sigma_\varepsilon^2}}{\sigma_v}. \quad (1)$$

Proof. Under the linear strategy $x = \beta(v - p_0)$ and price $p = p_0 + \lambda\tilde{y}$, the signal $\tilde{y}$ is jointly Gaussian with v :

$$\tilde{y} = \beta(v - p_0) + u + \varepsilon$$

has mean 0, variance $\beta^2\sigma_v^2 + \sigma_u^2 + \sigma_\varepsilon^2$, and covariance with v equal to $\text{Cov}(v, \tilde{y}) = \beta\sigma_v^2$. By the projection formula for jointly Gaussian variables,

$$\mathbb{E}[v \mid \tilde{y}] = p_0 + \frac{\beta\sigma_v^2}{\beta^2\sigma_v^2 + \sigma_u^2 + \sigma_\varepsilon^2} \cdot \tilde{y}. \quad (2)$$

The Bayesian-rationality condition matches this against $p_0 + \lambda\tilde{y}$, yielding

$$\lambda = \frac{\beta\sigma_v^2}{\beta^2\sigma_v^2 + \sigma_u^2 + \sigma_\varepsilon^2}. \quad (3)$$

For the informed trader's best response, condition on v and optimize over x given the MM's price schedule. Since $u, \varepsilon \perp\!\!\!\perp v$ implies $\mathbb{E}[u \mid v] = \mathbb{E}[\varepsilon \mid v] = 0$,

$$\begin{aligned}\mathbb{E}[(v - p(\tilde{y})) \cdot x \mid v] &= \mathbb{E}[(v - p_0 - \lambda(x + u + \varepsilon)) \cdot x \mid v] \\ &= (v - p_0)x - \lambda x^2.\end{aligned}$$

This is strictly concave in x with second derivative $-2\lambda < 0$. The unique maximizer is

$$x^* = \frac{v - p_0}{2\lambda},$$

so the best-response coefficient is $\beta = 1/(2\lambda)$.

Substituting $\beta = 1/(2\lambda)$ into (3) and simplifying:

$$\begin{aligned}\lambda \left(\frac{1}{4\lambda^2} \sigma_v^2 + \sigma_u^2 + \sigma_\varepsilon^2 \right) &= \frac{\sigma_v^2}{2\lambda}, \\ \frac{\sigma_v^2}{4\lambda} + \lambda(\sigma_u^2 + \sigma_\varepsilon^2) &= \frac{\sigma_v^2}{2\lambda}, \\ \lambda(\sigma_u^2 + \sigma_\varepsilon^2) &= \frac{\sigma_v^2}{4\lambda}, \\ \lambda^2 &= \frac{\sigma_v^2}{4(\sigma_u^2 + \sigma_\varepsilon^2)}.\end{aligned}$$

Taking the positive root (since $\lambda > 0$) yields $\lambda = \sigma_v / (2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2})$, and substituting back gives $\beta = 1/(2\lambda) = \sqrt{\sigma_u^2 + \sigma_\varepsilon^2} / \sigma_v$, establishing (1).

Uniqueness within the class of linear equilibria follows from strict concavity of the informed trader's objective (unique x^* given λ) together with the fact that substituting $\beta = 1/(2\lambda)$ into (3) reduces the system to $\lambda^2 = \sigma_v^2 / (4(\sigma_u^2 + \sigma_\varepsilon^2))$, which has a unique positive root. $\square$

Remark 1 (Sanity check: no-privacy limit). Setting $\sigma_\varepsilon = 0$ in (1) recovers the classical Kyle [11] equilibrium $\lambda = \sigma_v / (2\sigma_u)$, $\beta = \sigma_u / \sigma_v$. The committed Bayesian AMM coincides with the competitive zero-profit MM exactly in the no-privacy limit; the framing departure from textbook Kyle is inactive when $\sigma_\varepsilon = 0$.

Remark 2 (The subsidy is intrinsic to coarse-signal pricing).

The privacy subsidy is not an artifact of the committed-pricing label; it is the unavoidable cost of pricing on a signal strictly coarser than the executed flow. Two notions of zero profit must be separated. Zero profit *conditional on the maker's information* $\sigma(\tilde{y})$ forces $p = \mathbb{E}[v \mid \tilde{y}]$, which is at once the Bertrand-competitive price under coarsened observation *and* our committed Bayesian rule; since $y = x + u$ is not $\sigma(\tilde{y})$ -measurable when $\sigma_\varepsilon > 0$, this efficient price earns the negative profit $\mathbb{E}[(p - v)y] = -\lambda\sigma_\varepsilon^2 = -|\pi_M|$ on the real flow. Zero profit *unconditionally against the real flow* $\mathbb{E}[(v - p)y] = 0$ instead gives $\tilde{\lambda} = \sigma_v / (2\sigma_u)$, independent of σ_ε ; but this rule over-reacts to $\tilde{y}$ (its quote is not the posterior mean), so it is not implementable by a price-taking maker that observes only

$\tilde{y}$: a rival quoting the posterior would undercut it. Informational efficiency and zero profit against the real flow are therefore incompatible whenever $\sigma_\varepsilon > 0$, and the wedge between them is exactly the privacy subsidy $\lambda\sigma_\varepsilon^2$. The subsidy is the generic competitive outcome of coarse-signal pricing, not a consequence of relaxing zero profit; the Penumbra design of Section 6.2, where $\pi_M = 0$ because the maker prices on and trades against the *same* observable, is the converse — no coarsening, no subsidy.

4.1 Comparative statics

The equilibrium has clean comparative statics in σ_ε .

Proposition 1 (Comparative statics). *The equilibrium of Theorem 1 satisfies:*

1. $\partial\lambda/\partial\sigma_\varepsilon < 0$: the price-impact coefficient strictly decreases as privacy noise increases.
2. $\partial\beta/\partial\sigma_\varepsilon > 0$: the informed-trader strategy intensifies as privacy noise increases.
3. (Half-revealing identity.) $\lambda\beta = 1/2$ for all $\sigma_\varepsilon \geq 0$.

Proof. Direct differentiation of (1) gives (i) and (ii). For (iii),

$$\lambda\beta = \frac{\sigma_v}{2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2}} \cdot \frac{\sqrt{\sigma_u^2 + \sigma_\varepsilon^2}}{\sigma_v} = \frac{1}{2}$$

identically. □

The identity $\lambda\beta = 1/2$ holds in textbook Kyle, where it is the well-known half-revealing property: the equilibrium price moves halfway from the prior toward the true value, on average. Our result is that this identity *persists exactly* under privacy noise — a robustness statement about Kyle’s half-revealing property, not a novel quantity. Substituting the equilibrium back into $\tilde{y}$ yields

$$p = p_0 + \lambda\beta(v - p_0) + \lambda(u + \varepsilon) = \frac{p_0 + v}{2} + \lambda(u + \varepsilon).$$

The conditional mean $\mathbb{E}[p \mid v] = (p_0 + v)/2$ is independent of σ_ε ; only the conditional variance $\text{Var}(p \mid v) = \lambda^2(\sigma_u^2 + \sigma_\varepsilon^2) = \sigma_v^2/4$ is also independent of σ_ε . Both the information content and the realized noise of the price are preserved by privacy at the price level.

The cost of privacy is therefore not visible at the level of the price’s distribution conditional on v . As we show in Section 5, the cost appears instead in the expected P&L of the market maker against informed flow, and this is what the LP pool / protocol treasury must absorb.

5 Welfare decomposition: the privacy subsidy

This section computes the per-period expected profit or loss of each agent under the equilibrium of Theorem 1 and identifies the *privacy subsidy* as a closed-form quantity transferred from the protocol/LP pool to traders.

5.1 Per-agent expected P&L

The three relevant per-period expected quantities are:

$$\begin{aligned}\pi_I &:= \mathbb{E}[(v - p) \cdot x], & (\text{informed trader's expected profit}), \\ \pi_N &:= \mathbb{E}[(v - p) \cdot u], & (\text{noise traders' expected net P\&L, a loss}), \\ \pi_M &:= \mathbb{E}[(p - v) \cdot (x + u)], & (\text{MM/protocol expected profit on flow}).\end{aligned}$$

By construction $\pi_I + \pi_N + \pi_M = 0$: the three components form a zero-sum decomposition of the total trade-realized P&L against the asset's terminal value v .

Lemma 1 (Per-agent P&L formulas). *Under the equilibrium (β, λ) of Theorem 1:*

$$\pi_I = +\frac{1}{2} \sigma_v \sqrt{\sigma_u^2 + \sigma_\varepsilon^2}, \quad (4)$$

$$\pi_N = -\frac{\sigma_v \sigma_u^2}{2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2}}, \quad (5)$$

$$\pi_M = -\frac{\sigma_v \sigma_\varepsilon^2}{2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2}}. \quad (6)$$

Proof. Use $x = \beta(v - p_0)$, $u, \varepsilon \perp v$, and the half-revealing identity $\lambda\beta = 1/2$ from Proposition 1.

(i) *Informed.*

$$\begin{aligned}\pi_I &= \mathbb{E}[(v - p_0 - \lambda(x + u + \varepsilon)) \cdot \beta(v - p_0)] \\ &= \beta \mathbb{E}[(v - p_0)^2] - \lambda\beta \mathbb{E}[(x + u + \varepsilon)(v - p_0)] \\ &= \beta \sigma_v^2 - \frac{1}{2} \beta \sigma_v^2 = \frac{1}{2} \beta \sigma_v^2.\end{aligned}$$

Substituting $\beta = \sqrt{\sigma_u^2 + \sigma_\varepsilon^2}/\sigma_v$ yields (4).

(ii) *Noise.* Since $u \perp (x, \varepsilon)$ and $\mathbb{E}[u] = 0$,

$$\pi_N = \mathbb{E}[(v - p_0 - \lambda(x + u + \varepsilon)) \cdot u] = -\lambda \mathbb{E}[u(x + u + \varepsilon)] = -\lambda \sigma_u^2.$$

With $\lambda = \sigma_v/(2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2})$, this gives (5).

(iii) *MM/protocol.* By the zero-sum identity,

$$\pi_M = -(\pi_I + \pi_N) = -\frac{1}{2} \sigma_v \sqrt{\sigma_u^2 + \sigma_\varepsilon^2} + \frac{\sigma_v \sigma_u^2}{2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2}} = -\frac{\sigma_v \sigma_\varepsilon^2}{2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2}},$$

which is (6). □

5.2 The privacy subsidy

The main welfare result follows directly.

Theorem 2 (Privacy subsidy). *Under the equilibrium of Theorem 1, the protocol's expected loss per period is*

$$|\pi_M| = \frac{\sigma_v \sigma_\varepsilon^2}{2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2}} \geq 0, \quad (7)$$

with equality if and only if $\sigma_\varepsilon = 0$. The quantity $|\pi_M|$ is the privacy subsidy: the per-period transfer from the protocol/LP pool to traders, induced by the privacy mechanism. For protocol break-even, the total fees collected per period must satisfy

$$\text{fees}_{\text{period}} \geq |\pi_M|.$$

Proof. Equation (7) is (6) of Lemma 1; non-negativity is immediate. Equality at $\sigma_\varepsilon = 0$ recovers the textbook-Kyle MM zero-profit identity (Remark 1). $\square$

Naming and fee model. We label $|\pi_M|$ *privacy subsidy* as a memorable analog to LVR; equivalent neutral phrasings are *noise-induced LP loss* or *protocol's adverse-selection cost from privacy*. The break-even bound is computed against the no-fee equilibrium: a per-trade fee f breaks the linear-strategy structure (informed traders cease trading when $|v - p_0| < f$), so a complete fee-equilibrium analysis with endogenous volume response is left for future work.

5.3 Comparative statics of the subsidy

Proposition 2 (Subsidy asymptotics and shape). *Let $|\pi_M|(\sigma_\varepsilon)$ denote the subsidy of (7) as a function of $\sigma_\varepsilon \geq 0$ with σ_v, σ_u fixed.*

1. *Low-privacy expansion: as $\sigma_\varepsilon \downarrow 0$, $|\pi_M|(\sigma_\varepsilon) = \frac{\sigma_v}{2\sigma_u} \sigma_\varepsilon^2 + O(\sigma_\varepsilon^4)$.*
2. *High-privacy limit: as $\sigma_\varepsilon \rightarrow \infty$, $|\pi_M|(\sigma_\varepsilon) = \frac{1}{2} \sigma_v \sigma_\varepsilon + O(\sigma_\varepsilon^{-1})$.*
3. *$|\pi_M|$ is strictly increasing in σ_ε on $[0, \infty)$.*
4. *$|\pi_M|$ has a single inflection point at $\sigma_\varepsilon^* = \sqrt{2} \sigma_u$: it is convex on $[0, \sigma_\varepsilon^*]$ and concave on $[\sigma_\varepsilon^*, \infty)$.*

Proof (Proof of (iii)–(iv)). Differentiating $|\pi_M|$ twice in σ_ε ,

$$\frac{\partial |\pi_M|}{\partial \sigma_\varepsilon} = \frac{\sigma_v \sigma_\varepsilon (2\sigma_u^2 + \sigma_\varepsilon^2)}{2(\sigma_u^2 + \sigma_\varepsilon^2)^{3/2}} > 0$$

for $\sigma_\varepsilon > 0$, establishing (iii). The second derivative simplifies to

$$\frac{\partial^2 |\pi_M|}{\partial \sigma_\varepsilon^2} = \frac{\sigma_v \sigma_u^2 (2\sigma_u^2 - \sigma_\varepsilon^2)}{2(\sigma_u^2 + \sigma_\varepsilon^2)^{5/2}},$$

which vanishes exactly at $\sigma_\varepsilon^2 = 2\sigma_u^2$, is positive for $\sigma_\varepsilon^2 < 2\sigma_u^2$, and negative for $\sigma_\varepsilon^2 > 2\sigma_u^2$. This gives (iv). $\square$

Item (i): for small privacy noise, the subsidy is quadratic in σ_ε — doubling the privacy parameter quadruples the LP-pool cost. Item (ii): asymptotically linear growth in the high-privacy regime. Item (iv): the protocol's marginal cost $\partial |\pi_M| / \partial \sigma_\varepsilon$ is itself increasing for small σ_ε but eventually decreasing as σ_ε grows past $\sqrt{2} \sigma_u$ — past this threshold, additional privacy is *less* expensive at the margin. The inflection at $\sigma_\varepsilon^* = \sqrt{2} \sigma_u$ is a structural feature of the square-root denominator in (7).

5.4 Welfare incidence

Corollary 1 (Noise traders also benefit from privacy). $\partial\pi_N/\partial\sigma_\varepsilon > 0$ for all $\sigma_\varepsilon > 0$: the noise (uninformed) traders' expected loss is strictly decreasing in the privacy parameter.

Proof. Differentiating (5),

$$\frac{\partial\pi_N}{\partial\sigma_\varepsilon} = \frac{\sigma_v \sigma_u^2 \sigma_\varepsilon}{2(\sigma_u^2 + \sigma_\varepsilon^2)^{3/2}} > 0 \quad \text{for } \sigma_\varepsilon > 0. \square$$

Combined with Theorem 2, the welfare picture under privacy is therefore:

- informed trader gains: $\partial\pi_I/\partial\sigma_\varepsilon > 0$;
- noise trader gains (*loses less*): $\partial\pi_N/\partial\sigma_\varepsilon > 0$;
- protocol/LP pool loses by exactly that sum: $\partial|\pi_M|/\partial\sigma_\varepsilon > 0$;
- total welfare is preserved (zero sum).

Privacy improves π_N , but not at the informed trader's expense. Both trader types gain; the entire transfer is borne by the protocol. In the low-privacy regime, the leading-order Taylor expansions of π_I and π_N around $\sigma_\varepsilon = 0$ are equal,

$$\pi_I(\sigma_\varepsilon) - \pi_I(0) = \pi_N(\sigma_\varepsilon) - \pi_N(0) = \frac{\sigma_v}{4\sigma_u} \sigma_\varepsilon^2 + O(\sigma_\varepsilon^4),$$

so the privacy subsidy is split symmetrically between informed and noise traders to leading order. The asymmetry only appears at higher orders: in the high-privacy limit, the informed trader captures essentially all the subsidy ($\pi_I \sim \sigma_v \sigma_\varepsilon/2$) while the noise trader's loss vanishes ($\pi_N \rightarrow 0$, i.e. their gain over the classical Kyle benchmark saturates at $\sigma_v \sigma_u/2$).

Remark 3 (The privacy "gain" is gross-of-fees; Corollary 1 is welfare-neutral net-of-fees). The statement "noise traders also benefit from privacy" (Corollary 1) is a *gross-of-fees* observation about the no-fee equilibrium of Theorem 1. We now show that, at the same no-fee equilibrium volumes, charging the break-even fee of Theorem 2 via a volume-proportional levy exactly cancels both traders' incremental gains.

At equilibrium the expected absolute volumes are $\mathbb{E}|x| = (\sigma_v/2\lambda)\sqrt{2/\pi}$ and $\mathbb{E}|u| = \sigma_u\sqrt{2/\pi}$, so total volume $Q := \mathbb{E}|x| + \mathbb{E}|u| = \sqrt{2/\pi}(\sqrt{\sigma_u^2 + \sigma_\varepsilon^2} + \sigma_u)$ using $\lambda = \sigma_v/(2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2})$ from Theorem 1. The break-even rate is $f = |\pi_M|/Q$. Each side pays its volume share:

$$\begin{aligned} f \cdot \mathbb{E}|x| &= \frac{\sigma_v(\sqrt{\sigma_u^2 + \sigma_\varepsilon^2} - \sigma_u)}{2} = \pi_I(\sigma_\varepsilon) - \pi_I(0), \\ f \cdot \mathbb{E}|u| &= \frac{\sigma_v \sigma_u(\sqrt{\sigma_u^2 + \sigma_\varepsilon^2} - \sigma_u)}{2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2}} = \pi_N(\sigma_\varepsilon) - \pi_N(0). \end{aligned}$$

Each fee equals the corresponding trader's incremental gain over the $\sigma_\varepsilon = 0$ baseline; net-of-fees, both $\pi_I - f \mathbb{E}|x| = \sigma_v \sigma_u/2$ and $\pi_N - f \mathbb{E}|u| = -\sigma_v \sigma_u/2$

revert to the classical Kyle values, while the MM is exactly compensated. Privacy is therefore *exactly welfare-neutral* under the volume-proportional break-even fee, at the partial-equilibrium level of analysis (no-fee equilibrium volumes, with fee revenue redistributed back to the LP pool). The full fee-equilibrium analysis, in which the fee distorts the linear-strategy structure as flagged after Theorem 2, remains open and may yield additional deadweight loss.

6 Application to zero-knowledge exchange designs

The model of Section 3 applies cleanly to one class of zero-knowledge exchange designs: smart-contract AMMs whose observation of order flow is perturbed by an explicit additive Gaussian noise injection. We treat this canonical application in Section 6.1, then situate three nearby privacy designs that do *not* fit our framework as currently formulated in Section 6.2.

6.1 Primary application: shielded AMMs with DP-style noise

A *shielded AMM with differential-privacy (DP) noise* is a constant-function-style AMM augmented with a privacy layer that injects calibrated Gaussian noise into the observable order flow before the AMM applies its update rule. Concretely, after each trade the protocol observes $\tilde{y} = y + \varepsilon$ with $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ drawn by the privacy mechanism, and the AMM updates reserves — and therefore the spot price — according to this noisy signal. The privacy parameter σ_ε is exactly the standard deviation of the injected DP noise.

The mapping is literal: σ_ε in our framework equals the DP noise scale in the implementation. Theorem 2 therefore yields a direct break-even fee prescription for any such shielded AMM: per-period fees must total at least the subsidy $|\pi_M|$ to compensate the LP pool. Appendix A tabulates the resulting fee floor in USD per day under a BTC/USDT calibration; at $\sigma_\varepsilon = \sigma_u$, the LP/protocol must collect approximately \$1M/day — comparable to the entire revenue from a 0.1% fee on \$1B of daily volume.

This setting pairs cleanly with the DP-feasibility analysis of Chitra, Angeris, and Evans [7], who study a *Uniform Random Execution* mechanism that achieves (ε, δ) -DP in constant-function market makers and characterize the privacy parameter as a function of curvature and trade count. Their analysis is on the realizability side: when and how DP can be achieved in a CFMM. Our framework is the complementary welfare side: given that a DP layer of intensity σ_ε is in place, what does the LP pool pay?

6.2 Mechanisms outside our framework

Three nearby privacy designs warrant separate analysis frameworks that our additive-Gaussian-noise model does not capture.

Penumbra-style batched swaps. Penumbra [14] clears swaps via per-block batches in which the cleared price depends on the aggregate batch flow. The Bayesian-MM observation is the *exact* aggregate $Y_\tau = \sum_t (x_t + u_t)$ over τ pooled noise-trader draws — not a noise-perturbed version of Y_τ . Treating one batch as one period (so per-batch noise-trader variance is $\tau\sigma_u^2$), direct calculation gives $\lambda = \sigma_v/(2\sigma_u\sqrt{\tau})$ and the informed batch-total coefficient $\beta = \sigma_u\sqrt{\tau}/\sigma_v$ — textbook Kyle with the rescaling $\sigma_u \rightarrow \sigma_u\sqrt{\tau}$. Because the MM both prices on and trades against the same observable Y_τ , the Bayesian projection identity gives $\pi_M = 0$ exactly. Batching reshapes market depth and informed-trader intensity but generates no privacy subsidy under this framework.

Suave-style sealed-bid order-flow auctions. Sealed bidding with delayed reveal [8] creates *temporal* information asymmetry rather than additive observation noise. The block proposer commits to a price ex ante and trades against revealed flow ex post; the adverse-selection cost in this design is closer to LVR’s stale-price arbitrage (Section 7) than to our σ_ε -noise setting. A continuous-time analysis with explicit time-lag is required.

Renegade-style midpoint-pegged crossing. Renegade [2] matches peer orders at an external lit-exchange midpoint via MPC: the on-chain mechanism consumes no flow signal to compute price. Such oracle-pegged designs fall entirely outside any Kyle-style analysis, since price discovery is exogenous.

7 Connection to Loss-Versus-Rebalancing

The privacy subsidy of Theorem 2 occupies the same conceptual slot as the *Loss-Versus-Rebalancing* (LVR) measure of Milionis et al. [13]. Both are closed-form, per-period welfare quantities that measure the adverse-selection cost borne by an automated pricing mechanism in the presence of informed traders, and both yield direct break-even fee prescriptions.

The source of the cost differs across the two results:

- LVR captures the cost of *stale prices*: the AMM’s price function is committed and lags the true reference price which arbitrageurs can read externally. The information asymmetry is temporal.
- The privacy subsidy captures the cost of *privacy noise*: the AMM observes its price-relevant signal with additive Gaussian noise injected by the privacy mechanism. The information asymmetry is informational, not temporal.

A natural open question is whether these costs combine additively in a continuous-time AMM that is simultaneously price-lagged and privacy-aggregated. A formal model would require lifting LVR’s continuous-time setup and our single-period Kyle setup into a common framework, which we do not attempt here. As a heuristic suggestion for follow-up work, we conjecture that the break-even fee in such a setting decomposes to leading order as

$$\text{fees}_{\text{period}} \geq \text{LVR} + |\pi_M|,$$

but emphasize that this is speculative. The cross-terms between time-lag and privacy noise need not vanish at higher orders: an arbitrageur exploiting a stale price may also exploit the privacy-noise statistics, and these channels' welfare costs need not be independent. A vanishing of cross-terms at leading order would require ε to be temporally uncorrelated with the stale-price process. A rigorous treatment is left for future work.

8 Conclusion

We have derived the closed-form linear Kyle equilibrium in a single-period market in which a committed Bayesian AMM observes order flow perturbed by independent Gaussian privacy noise of scale σ_ε . The equilibrium price-impact coefficient is $\lambda = \sigma_v / (2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2})$, monotonically decreasing in σ_ε . The corresponding informed-trader strategy intensifies symmetrically as $\beta = \sqrt{\sigma_u^2 + \sigma_\varepsilon^2} / \sigma_v$, and the product $\lambda\beta = 1/2$ is invariant in σ_ε . The welfare decomposition identifies a per-period transfer $|\pi_M| = \sigma_v \sigma_\varepsilon^2 / (2\sqrt{\sigma_u^2 + \sigma_\varepsilon^2})$ from the protocol/LP pool to traders. This is the *privacy subsidy*, the break-even fee that any privacy-aggregated exchange must charge to compensate its liquidity layer. The result applies directly to shielded AMMs with additive-noise (differential-privacy) injection. Related privacy designs — Penumbra-style batched swaps, Suave-style sealed-bid order-flow auctions, and oracle-pegged crossings such as Renegade — require separate frameworks: batching reduces to Kyle with rescaled noise-trader variance (no subsidy under Bayesian MM); sealed-bid auctions sit closer to LVR's temporal-asymmetry regime; oracle-pegged designs fall entirely outside Kyle.

Future work. Extensions left to subsequent work include a Glosten–Milgrom bid-ask-spread analog under privacy noise; other privacy mechanisms (directional-only, bucketed, time-delayed observation); multi-period dynamics aimed at a combined LVR-plus-privacy decomposition; mechanized formalization in a proof assistant; and empirical calibration of σ_ε from live zero-knowledge exchange traces.

A Numerical illustration

The closed-form results of Theorems 1 and 2 admit immediate numerical evaluation. Figure 1 plots $|\pi_M|(\sigma_\varepsilon)$ for $\sigma_v = \sigma_u = 1$ over $\sigma_\varepsilon \in [0, 5]$; Tables 1 and 2 tabulate λ , β , and $|\pi_M|$ at representative parameter values.

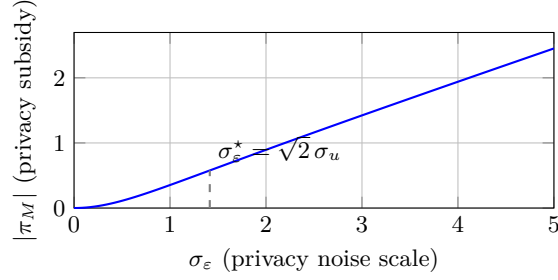


Fig. 1. Privacy subsidy $|\pi_M|$ vs. σ_ε for $\sigma_v = \sigma_u = 1$. The convexity-to-concavity inflection at $\sigma_\varepsilon^* = \sqrt{2}\sigma_u$ marks the transition between the quadratic low-privacy regime and the linear high-privacy regime.

A.1 Dimensionless table

Table 1 reports λ , β , and $|\pi_M|$ in units where $\sigma_v = \sigma_u = 1$. The convexity-to-concavity inflection of $|\pi_M|$ at $\sigma_\varepsilon^* = \sqrt{2}\sigma_u \approx 1.414$ (Proposition 2(iv)) is bracketed by the $\sigma_\varepsilon \in \{1.0, 1.414, 2.0\}$ rows.

σ_ε	λ	β	$ \pi_M $	note
0	0.500	1.000	0.000	textbook Kyle
0.5	0.447	1.118	0.112	low-privacy regime
1.0	0.354	1.414	0.354	
$\sqrt{2}$	0.289	1.732	0.577	$\sigma_\varepsilon = \sigma_\varepsilon^*$ (inflection)
2.0	0.224	2.236	0.894	past inflection
3.0	0.158	3.162	1.423	high-privacy
5.0	0.098	5.099	2.451	far high-privacy

Table 1. Dimensionless equilibrium values for $\sigma_v = \sigma_u = 1$, varying σ_ε . The privacy subsidy $|\pi_M|$ grows quadratically near $\sigma_\varepsilon = 0$ (slope $\rightarrow 0$), becomes maximally steep near $\sigma_\varepsilon^* = \sqrt{2}\sigma_u$, and asymptotes to linear $|\pi_M| \approx \sigma_\varepsilon/2$ for large σ_ε .

A.2 BTC-calibrated example

For an illustrative BTC/USDT calibration, take a per-day window with $\sigma_v = \$3,000$ (corresponding to $\sim 3\%$ daily volatility on a $\$100,000$ asset) and $\sigma_u = 1,000$ BTC per day. Table 2 reports $|\pi_M|$ in USD per day across several σ_ε values, expressed as a multiple of σ_u .

$\sigma_\varepsilon/\sigma_u$	$ \pi_M $ (USD/day)	As fraction of $\sigma_v\sigma_u$
0.1	\$~ 15,000	0.0050
0.5	\$~ 335,000	0.112
1.0	\$~ 1,060,000	0.354
$\sqrt{2}$	\$~ 1,730,000	0.577
2.0	\$~ 2,680,000	0.894

Table 2. Per-day privacy subsidy in USD for a BTC/USDT calibration with $\sigma_v = \$3,000/\text{day}$ and $\sigma_u = 1,000 \text{ BTC}/\text{day}$. (σ_u is the noise-trader-component standard deviation, not total daily volume; typical BTC/USDT daily volume is one to two orders of magnitude larger.) At $\sigma_\varepsilon = \sigma_u$, the LP/protocol must collect approximately \$1M/day in fees to break even.

For context, BTC/USDT spot volume on a typical centralized exchange is on the order of \$1–\$5 billion per day. A 0.1% fee on \$1B daily volume yields \$1M/day in revenue. Under the calibration of Table 2, this revenue roughly breaks even against the privacy subsidy at $\sigma_\varepsilon = \sigma_u$ (the subsidy is $\sim \$1.06\text{M}$, so a 0.1% fee on \$1B falls about 6% short), and falls further short for larger σ_ε . The implication for privacy-aggregated exchange designers is that fee schedules and privacy parameters cannot be chosen independently.

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